

جامعة أم القرى
UMM AL-QURA UNIVERSITY

ISA theoretical project report

Group 8
instructor Name : Ahmed Morgan

group 8 Laptops Comparison

Name in Arabic	سحر هوساوي	رفاة الشريف	ليان الصبحي
Student ID	446002564	446002882	445004505
Test Link	https://browser.geekbench.com/ai/v1/506024	https://browser.geekbench.com/ai/v1/506014	https://browser.geekbench.com/ai/v1/490937
User	SMhawsawi	ryalsharif	Lmalsubhi
Upload Date	May 25 2026 12:49 AM	May 24 2026 11:41 PM	April 28 2026 08:18 AM
Single precision Score (SP)	1986	1794	778
Half precision Score (HP)	865	918	406
Quantized Score (Q)	3987	3158	1070
ISA	x86-64	x86-64	x86-64
Frame Work	ONNX	ONNX	ONNX
Operating System	Microsoft Windows 11 Home (64-bit)	Microsoft Windows 11 Home (64-bit)	Microsoft Windows 11 Home (64-bit)
Processor Name	Intel Core i7-1360P @ 2.20 GHz	AMD Ryzen 5 7535HS @ 3.30 GHz	Intel Core i5-8250U @ 1.80 GHz
topology	1 Processor, 12 Cores, 16 Threads	1 Processor, 6 Cores, 12 Threads	1 Processor, 4 Cores, 8 Threads
Memory Size	16.00 GB	8.00 GB	8.00 GB
Workload (SP)	Style Transfer : 8004 (10.3 IPS)	Style Transfer : 6817 (IPS 8.76)	Style Transfer: 2665 (3.43 IPS)
Workload (HP)	Style Transfer : 5769 (7.42 IPS)	Style Transfer :I 5813 (7.47 IPS)	Style Transfer: 1928 (2.48 IPS)
Workload (Q)	Style Transfer : 14429 (18.6 IPS)	Style Transfer : 10283 (13.4 IPS)	Style Transfer: 3968 (5.16 IPS)

Result

Based on the Geekbench AI results in the table, Sahar Hawsawi's laptop (Intel Core i7-1360P) is the strongest overall system, followed by Rafah Al Sharif's Ryzen 5 7535HS, while Layan Al-Subhi's i5-8250U comes last.

performance improves when the processor can execute instructions with fewer clock cycles (lower CPI) or process more instructions simultaneously. The Intel Core i7-1360P benefits from a newer architecture and 12 cores / 16 threads, allowing it to execute a larger number of instructions concurrently than the competing processors.

increasing the number of processing cores improves performance for workloads that can be executed in parallel. The i7-1360P has the highest processing resources, enabling better throughput during multitasking operations.

Style Transfer is an artificial intelligence technique that applies the visual style of one image (such as a painting or sketch) to another image while preserving the original content.

Based on the formula: Performance = 1 / Execution Time
The device processing more images per second (higher IPS) has a lower execution time, and is therefore faster. Saher's device achieved 18.6 IPS in Quantized, compared to 13.4 for Rafah and 5.16 for Layan — meaning the Saher device is 3.6× faster than Layan. Also, based on: CPU Time = Instruction Count × CPI × Clock Cycle Time
The i7-1360P processor features 12 cores and 16 threads, allowing it to execute more instructions simultaneously, which directly reduces total execution time.

Q Workload	HP Workload	SP Workload	Processor
14429	5769	**8004**	Intel Core i7-1360P
10283	**5813**	6817	AMD Ryzen 5 7535HS
3968	1928	2665	Intel Core i5-8250U

Although the AMD Ryzen 5 7535HS achieved a slightly higher Half Precision workload score (5813 versus 5769), the difference is very small (less than 1%).